Multi-path Video Streaming in Vehicular Communication: Approaches and Challenges.

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***Abstract*—Vehicular Ad-hoc Networks (VANETs) has witnessed significant attention from academia and industries, due to the growing number of applications and devices focusing on safety and efficiency in transportation. Video Streaming in Vehicular Communication (VSVC) presents interactive real-time view of an emergency incidence on roads. On-board unit of vehicles are utilized for streaming real time video of on road traffic, and disseminate among vehicles via Vehicle to Vehicle (V2V) and Vehicle to Infrastructure (V2I) communication. VSVC is a challenging task due to the delay constrained nature of video data and dynamic vehicular network environment. Some of the fundamental issues in video streaming include queueing delay, bandwidth underutilization, and intermittent connectivity. Various techniques have been suggested in literature to address the issues in video streaming. To this end, this paper presents a qualitative review on multipath video streaming focusing on vehicular communication. Specifically, review of different multipath approaches with their comparative assessment are presented. Major challenges in multipath video streaming are identified as future directions of research. The review would benefit researchers and practitioners, in terms of enhancing clarity in multipath video streaming literature and emphasizing the issues.**

Keywords—***Cross-layer; Video streaming; Multimedia, Compression; Vehicular communication***.

I. INTRODUCTION

In recent time, Mobile Ad hoc Networks (MANETs) has introduced a new paradigm called VANETs [1]. In VANETs, vehicles act as a network nodes with higher mobility during communication. It generally comprises Roadside Units (RSUs), OBUs and in-built sensors. RSUs are normally installed beside the roads in highway and urban environments. RSUs facilitate vehicular communication in two modes including V2V and V2I. The OBUs are often fitted inside vehicles, for example Dedicated Short Range Communication (DSRC) devices which supports short distant communication over V2V and V2I. The vehicular communication paradigm supports cooperative communication and networking for ITS [2]. ITS defines various applications including emergency brake cautioning, cooperative navigation control, security distance cautioning, driver support, collision cautioning and supportive driving [3]. Vehicles as network node in VANETs do not have the battery power issue, because of its high capacity automatic rechargeable battery. The batteries enables intensive and rigorous data communications and processing capability. The intelligent communication gadgets including smart antenna, Global Positioning System (GPS) and smart onboard buffer storage supports the abilities to achieve efficient video streaming forwarding and cooperative downloading over V2V and V2I [4].

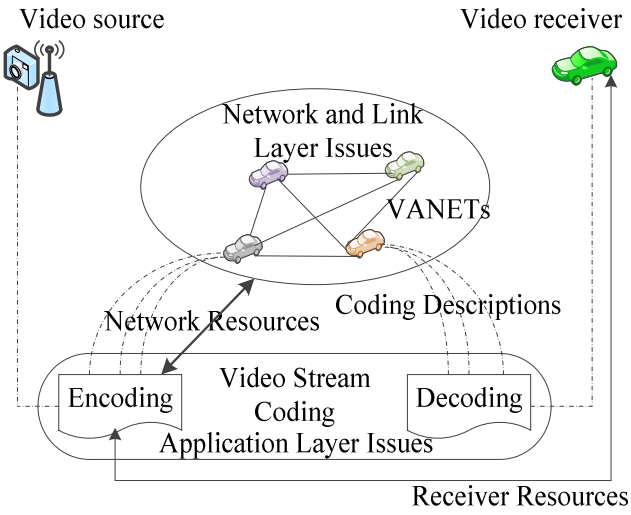


Fig. 1. Video streaming in vehicular communication

Video streaming is performed, when video content is transmitted in encoded form over the network, and displays the content to the end user in a pre-recorded or recorded manner [5]. The advantage of efficient streaming is that the user does not need to waste time in downloading a video content for playing the video. Instead, the media performs the task of sending continuous streams of data pre-recoded video playing [6]. In current approaches, video streams are usually compressed and partitioned into sub-streams, to reduce video size and achieve efficient load balancing in video streaming [7]. Most commonly used compression standard includes MPEG-1-2-4, H.264/AVC, and H.263/SVC [8, 9]. The compression standard are used to reduce redundant and irrelevant video data. The redundancy could be color space, temporal or spatial, while the irrelevant information in video data is considered unimportant information [9]. The encoding process is carried out at source node before sending as video stream, while decoding is performed at the destination after delivery, i.e., before viewing the video content. The video partitioning leverages encoding methods where video streams are coded

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frames including I-frames, P-frames and B-frames.

The different coded frames sub-streams can be transmitted using single and multipath communication. Figure 1 illustrate video streaming in terms of network protocol stack. Video streaming in vehicular environment is a challenging task due to the dynamic network topology and delay constrained nature of video data. A resilient video streaming must be capable of withstanding deep connection fading and connection failures. Intermittent disconnections due to the high mobility of vehicles leads to packets loss in video streaming. Further, bandwidth underutilization causes packet congestion, packet burst, and packet collision. In addition, the lack of effective priority path selection techniques for video streaming has brought about the issue of queuing delay. Several techniques are suggested for addressing the aforementioned challenges.

In this paper, we have critically reviewed different multipath approaches and techniques. We present a comparative assessment table for multipath approaches used in recent work. Further, we highlighted some open challenges and issues in video streaming in vehicular communication (VSVC).

II. Multipath Video Streaming

In this section, we present a description of multipath connectivity graph and a review on multipath transmission strategy in video streaming in vehicular communication.

In a multipath connectivity graph, a VANET is created by a set of N nodes (vehicles) where N = 1, … , n and each vehicle is equipped with a single radio interface. The channels available in the network is denoted as C, where C = 1, …, c and C_{Max} is the highest bandwidth of each channel. The IEEE 802.11p/WAVE protocol provides 1–4Mbps for Japan, 250Kbps for Europe and 3–27Mbps for USA. All nodes are assumed to be dynamic with a variable speed. Further, nodes operate with different transmission power thus, having creating different transmission coverages. Therefore, a link l between two nodes is active if and only if, it is functioning on a single channel. The link l_1 is interfered if there exist a certain coverage area of collision domain where another link l_2 lies in the same channel assigned to l_1. The interfered area is a shared physical coverage area of transmission area of sender and receiver.

A graph G consist of points and edges (P, E). The elements of P are called nodes and the elements of E are the connecting links between points of the graph. We consider vehicular network topology as a dynamic graph structure. Let G be a graph with set of paths M. In general, a path in a graph constitute of series of distinct points p that is, set of nodes $p_1, p_2, p_3, \ldots, p_k$. $\exists\, pipi+1$ is an edge E, that is link between two different points, $\forall\, i = 1, \ldots, k\text{-}1$. The length L of single path M is the sum of all Es in a single M. Hence, we infer that the angle θ between selected multiple paths can be directly proportional to the interference coverage of each p in the paths M.

$$M\theta\ \alpha\ CM \qquad (1)$$

Therefore, if the angle θ is assumed and calculated before transmitting a video stream through the selected path considering line of direction of relaying node. Hence, interference between multiple paths could be reduced or even avoided and qualitative video streaming delivery can be achieved.

Video streaming in vehicular communication (VSVC) is qualitatively reviewed on the basis of multipath strategy. The approaches include Media Access Control (MAC), Forward error (FEC), Routing-centric Coding (RcC), Quality of Service aware (QoS), Quality of Experience aware (QoE) and Cross layer (CL) approaches. The objective approach are all based on single objective. The metrics include Delay (DL), Delivery Ratio (DR), Throughput (THP), SSIM (Structural Similarity index (SSIM), Mean Opinion Score (MOS), Peak Signal to Noise Ratio (PSNR), jitter (JTR), Control Overhead (CO), Frame Loss (FL) and COST as seen in Table 1. Despite many solutions proposed, there is need to improve on collision problem which occur during transmission through multiple path selection. That is why, we have performed a critical review in this regard.

Multipath video streaming in vehicular communication is an approach which forwards sub-streams through different paths from sender to the receiver vehicle. Multipath coding- centric routing consider the compression of video while at the same time choosing most suitable and reliable paths for video stream forwarding. In multipath video scheme, video streaming flow is partitioned into distinct paths during transmission. It reduces the high video data rate, by achieving load balancing during video transmission. Multipath video streaming mainly focus on path selection algorithm. It usually employs link/node disjoint approaches for efficient routing in video streaming. The multipath support the attainment of QoS in the following ways including fault tolerance, load balancing and, bandwidth and delay aggregation.

In [10], an approach which is based on cross-layer is presented to reduce video distortion in order to attain qualitative video delivery. It consider path selection techniques (CPSV). The cross-layer based approach maps application layer requirement by using routing layer to achieve its objectives. The approach considers coding rate allocation, it works in such a way that the transmission rate must not surpass the encoding rate of the network. However, An Application based V2V Routing Protocol for video streaming in multi-hop urban VANETs settings (AVRP) is presented [11]. The protocol is a framework for pre-recorded video dissemination over more than one hop in urban settings. The urban settings considered in designing the routing protocol include spatial traffic arrangement, queuing based movement model and probability of connectivity for fewer nodes and plenty nodes network.

Location-centric multipath approach for streaming video over VANETs (LIAITHON) is proposed to avoid route coupling effect [12]. The approach is centered on consideration of location parameters to select the best multiple paths for video stream forwarding. Further, it uses forwarding zone scheme for reducing collision and congestion problem. Furthermore, a Cluster-based solution for Multimedia Content Delivery (CMCD) is suggested to handle socio-economic issues in infotainment deployment in VANETs [13]. The suggested user-centric cluster based solution has the ability to handle vehicle passenger preferences and transmit multimedia content of either

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drivers' or passengers' interest. A K-hop video stream Forwarding scheme for Cooperative (KFC) vehicular network is suggested for alleviating bandwidth constraints [14]. The cooperative scheme works in such a way that a vehicle can request for assistance from other vehicles which are in the same fleet to help in downloading video content. A resilient location-oriented video unicast scheme for VANETs video streaming is proposed to handle VANETs dispersion and mobility issues [15]. The scheme employs VIdeo Reactive Tracking-based Unicast protocol (VIRTUS1). VIRTUS1 is a delivery-based solution that employs vehicles' existing and prospective position for selection strategy of relaying nodes. An Opportunistic Routing solution for pre- recorded Video (ORV) streaming is proposed for handling interference in wireless fading channels [16]. The scheme takes into consideration of the interference of vehicles from surroundings during the relay selection procedures. A multiple path solution with error correction for video streaming over VANETs (LIAITHON+) is presented [17]. The LIAITHON+ employs the multipath approach to distribute the high data rate traffic into a set of paths. Rezende, Mammeri [18] proposed a receiver-based video transmission solution for VANETs which has network with content transmission decoupled from relay node selection (REDEC).

De Felice, Cerqueira [19] suggested a non-centralized beaconless routing protocol for pre-recorded video transmission (DBD) over VANETs. It is an integrated framework that handles QoE of video services and routing protocol. A Learning Automata (LA) with Bayesian Coalition Game Theory (BCGT) is used to design an intelligent Mobile Video Surveillance System (MVSS) (BLVS) [20]. The LA is considered to be the players in the game and they also form coalition among themselves by using the BCGT. Al-Ani and Seitz [21] presents a video stream routing QoS scheme for multi-rate mechanism, in order to achieve congestion control and avoidance. The mechanism employs Ant Colony Optimization (ACO) algorithm and Simple Network Management Protocol (SNMP) monitoring features. The mechanism is called QoRA. Table I shows comparative assessment of multipath routing techniques for VSVC.

A technique termed SHIELD for FEC has been suggested in [22]. It is based on minimizing packet loss in pre-recoded video streaming over VANETs. Furthermore, an Adaptive Video Streaming System over a Cooperative (AViSCo) fleet of vehicles using the mobile bandwidth aggregation approach is suggested. [23]. Precisely, it is based on k-hop adaptive video transmission protocol to attain better throughput in video streaming over VANETs. Next is Table I, which depicts summary of multipath video streaming techniques.

TABLE I. SUMMARY OF THE RELATED LITERATURES ON MULTIPATH CODING-CENTRIC ROUTING VIDEO STREAMING OVER VANETS.

Protocol/Author	**Approach**						**Objective approach**		**Metrics**									
	MAC	***FEC***	***RcC***	***QoS***	***QoE***	***CL***	***SO***	***MO***	***DL***	***DR***	***THP***	***SSIM***	***MOS***	***PSNR***	***JTR***	***CO***	***FL***	***COST***
CPSV [10]		✓	✓			✓	✓							✓				
AVRP [11]			✓		✓												✓	
LIAITHON [12]			✓				✓		✓								✓	
CMCD [13]			✓			✓	✓				✓						✓	
KFC [14]			✓								✓							
VIRTUS1 [15]			✓				✓		✓								✓	✓
ORV [16]	✓		✓						✓					✓				
LIAITHON+[17]		✓	✓						✓								✓	✓
REDEC [18]			✓				✓		✓	✓				✓	✓			
DBD [19]			✓		✓				✓			✓	✓	✓				
BLVS [20]			✓						✓	✓	✓					✓	✓	
QoRA [21]			✓	✓			✓		✓	✓	✓				✓	✓		
SHIELD [22]	✓	✓			✓					✓		✓				✓		
AViSCO [23]			✓	✓							✓			✓		✓		

III. OPEN RESEARCH ISSUES AND CHALLENGES

The issues related to video streaming including strict requirements for lower delay, proper bandwidth utilization, jitter, packets loss, distortion etc. On the other hand the issues are related to wireless medium channels, the nature of the VANETs and the capabilities of transmitting nodes. Likewise in the case of multipath strategy, there is challenges of route coupling effect, if multiple path are not properly selected. Although, the overall aim is to attain acceptable perceived video quality. The conventional challenges of MANETs might not be directly the same as that of VANETs. Hence, we highlight the following as some of the major issues in multipath VSVC.

- Medium access control and, Error Correction and Recovery: The challenges is that, link layer approach requires so many adjustment of communication channels in order to suit VANETs nature. For example, vehicle movement pattern, driver behavior, and road topology need to be carefully modelled in order to get good signal reception and in turn achieving qualitative video stream delivery. The error correction and recovery issues in MANETs are strictly related to that of VANETs, except for the requirements of higher mobility.

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- Video coding over VANETs: Compression techniques which adequately considers VANETs features are required for better video streaming such as H.264/AVC. A more robust and flexible coding approaches that guarantees video stream delivery considering vehicle dynamicity need to be explored.
- Routing over VANETs: The challenges of path selection for video stream forwarding in VANETs are based on nature of wireless communication channels, properties of video and nature of the VANETs. There is need to consider conflicting multiple objectives hence, it becomes a challenging task. The multipath is employed in order to forward sub-streams, by this doing, there is possibility of packet loss. Therefore, a multipath approach that take into consideration the idea of disjointed nodes and paths is very important in order to achieve less delay in forwarding video streams.
- Overlay solution: this technique requires the creation of virtual replica of the main physical network in order to achieve decoupling, virtualization and easy management of the network nodes. The growth of virtual network cannot be translated directly to the physical network, since the virtual network is not real in implementation. Thus, only few nodes can be accommodated in an overlay solution.
- Cross-layer design in VANETs: the idea of cross layer design almost cut-across every proposal in both VANETs and MANETs. It is a widely accepted concept, because it improves video streaming concept considering the network protocol stack. The network protocol stack layers usually involved in cross-layering include *application, routing* and *link layer*. However, QoS provisioning is very challenging in cross layer approach. It is challenging because of provisioning of resources via all layers and the need for cross layer interaction.

IV. CONCLUSION

In this work, a comprehensive and descriptive review of related literature on multipath video streaming in vehicular communication (VSVC) is presented. Different approaches based on techniques employed in multipath video streaming are analyzed. Further, we have presented a comparative assessment table for the multipath approaches. We have also elaborate and condense some future research directions in video streaming. In conclusion, this review work will improve on the understanding of video streaming research trends and directions. Thus, this will stimulate research work in video streaming domain by making research gap finding easier. Consequently, better video streaming solutions will improve on-road infotainment and enhance on-road safety applications in vehicular communication.

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